

DATA MANAGEMENT

1. Policy and Practice

We will ensure that the data obtained and this study will be made available to the research community. The Bourns College of Engineering (BCOE) at UCR, in partnership with the UCR Libraries and the California Digital Library, has established a new, custom data management system designed to make results of our research readily and reliably accessible. This system combines two services of the California Digital Library (CDL) for the first time: an eScholarship series, a curated site where all BCOE investigators can store and disseminate their data, and EZID permanent identifiers for long-term identification and management of data resources.

All principal investigators in BCOE are responsible for complying with sponsor policies regarding data management and accessibility. All have sufficient computing resources to store and process the data generated in their research. Every principal investigator has the ability to create his or her personal web site, and to make data sets available through that web site. Additionally, the eScholarship series (<http://escholarship.org/uc/search?entity=bcoe>) provides another venue for dissemination. Researchers can submit materials to the eScholarship series site, where it is automatically given an EZID persistent identifier. All material on the site is indexed on Google, Google Scholar, and Melvyl/WorldCat. CDL also provides data backup services to avoid accidental loss of data. This partnership, which has resulted in the development of a new service that CDL can provide to other research organizations, makes our data widely available and searchable under the taxonomies, identifiers, and standards of universal data sharing systems envisioned for the near future (see *Changing the Conduct of Science in the Information Age*, NSF, June 2011).

2. Scope

This Data Management Plan addresses the NSF policy that primary data “commonly accepted in the scientific community as necessary to validate research findings” be made available at little or no cost to the PI or project. In accordance with this policy and guidance from the Office of Management and Budget, this plan does not include preliminary analyses (including raw data), drafts of scientific papers, plans for future research, peer reviews, or communications with colleagues. Data that must be withheld long enough to enable peer review and publication/dissemination or protection of intellectual property is subject to this plan only after those steps have taken place. Under this application, we will determine the effectiveness the proposed nano-vesicles (NETs) as photo-theranostic materials for fluorescence imaging and photothermal therapies.

The release of data about students or any other human subjects is subject to policies and restrictions in protocols adopted by the relevant Institutional Research Board (IRB) and the Family Educational Rights and Privacy Act (FERPA) regulations.

3. Data and Metadata Format and Contents

We will make the proposed nano-vesicles (NETs) available to interested investigators who would like to use them in their scientific research. We will publish our results, and present them results in relevant conferences, workshops, seminars, and teaching forums. Additionally, we will provide our results to any national resources and or databases on nanoparticles that compile and maintain such information. As an example, we will contribute to the nanotechnology information library (NIL) (<http://nanoparticlelibrary.net/index.asp>), which is maintained by the National Institute for Occupational Safety and Health (NIOSH). Data that must be withheld long enough to enable peer review and publication/dissemination or protection of intellectual property will be subject after appropriate steps have taken place. Confidential material will be handled according to policies and protocols at UCR, Family Educational Rights and Privacy Act (FERPA), and any other applicable regulations and restrictions.

All materials uploaded to the eScholarship site will be tagged uniformly with metadata about the source and nature of the data. Any electronic file format (e.g., text, spreadsheets, images) can be accepted. As a

service to assure that materials are formatted correctly, a curator from the College reviews all submissions before they are accepted.

4. Accessibility and Data Protection

Materials will be searchable through a number of databases and search engines, including Google and Google Scholar. As noted in Section 2 (Scope), data will be made available only after appropriate steps have been taken to protect intellectual property. Confidential material will be handled according to policies and protocols for human subjects, FERPA, and any other applicable regulations and restrictions.

Stored materials are downloadable but not editable, and they are backed up to assure no loss of data. All uploads are approved by a curator before they are published.

5. Derivative Products

All materials will contain acknowledgement of NSF support as per NSF policy: *“This material is based upon work supported by the National Science Foundation under Grant No. (NSF grant number).”* Additionally, materials will contain a disclaimer that *“any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.”* In keeping with standard ethical practices, it is expected that subsequent users of the data will acknowledge the source.

6. Archiving and Access

It is understood that the NSF Engineering Directorate requires data to be made available for a minimum of three years after conclusion of the award or public release, whichever is later. If the PI leaves his institution, the products will remain stored in relevant databases. PI is free to reproduce and repost these materials at new institution.